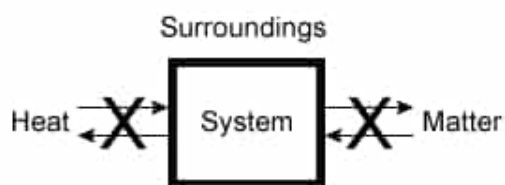


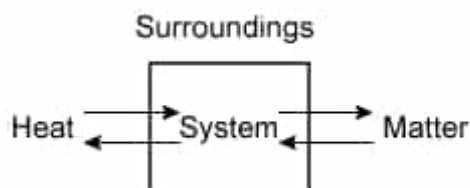
Which of the following diagrams illustrates the flow of heat and matter for a closed system?

☐ A.



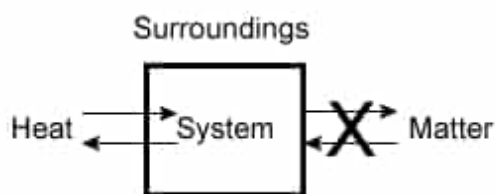
[18%]

☐ B.



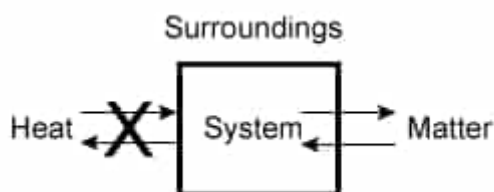
[1%]

☒ C.



[76%]

☐ D.



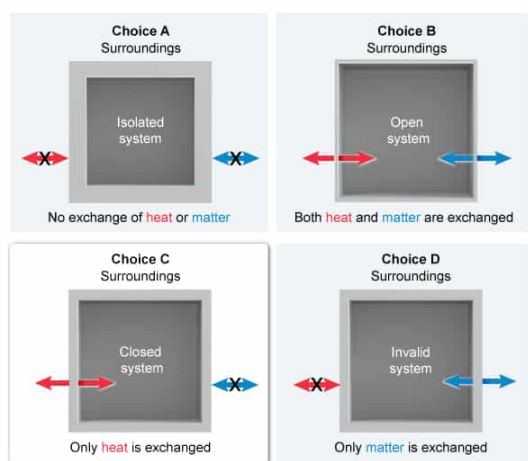
[3%]

Correct

76% answered correctly

Explanation:

Identifying a closed thermodynamic system



A **thermodynamic system** can be defined as a space chosen for study that is physically separated by a boundary (eg, surface of a wall) from its **surroundings** (ie, the rest of the universe). For example, during a chemical reaction, the reacting molecules are considered the system while all things not directly involved in the reaction (eg, solvent, reaction vessel) are considered the surroundings.

The extent to which heat and matter can be transferred between a system and its surroundings depends on the type of separation barrier. As such, there are three common types of thermodynamic systems.

- **Open system:** Both heat and matter are freely exchanged with the surroundings.
- **Closed system:** Only heat is exchanged with the surroundings.
- **Isolated system:** Neither heat nor matter can be exchanged with the surroundings.

In this question, the diagram presented in Choice C shows that **only heat** is entering and exiting the system. Based on the definitions given above, this diagram is best characterized as a **closed system**.

(Choice A) In this diagram, both sets of arrows never enter the system (ie, heat and matter are not exchanged); consequently, this is an *isolated* system.

(Choice B) Because both matter and heat are shown entering and exiting the system, this is an *open* system.

(Choice D) This diagram shows *only matter* entering and exiting the system. Therefore, this system cannot be defined as closed.

Educational objective:

A closed system interacts with its surroundings through the transfer of heat but not of matter.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403141

System:
Thermodynamics, Kinetics & Gas Laws